SET C

SUMMATIVE ASSESSMENT - III - 2025-26
SSLC Model Question Paper
Biology

Std: X

Max Score: 40
Time: 1.30 hrs.

Instructions

1. The first 15 minutes are cool-off time, to read and plan the answers.
2. Read all questions and instructions carefully before answering.
3. Consider score and time while writing answers.
4. Questions 5, 10, 13, 16, and 18 carry choices.

Answer questions from 1 to 4 each carries one score **(4×1=4)**

1. Which human ancestor among the following is the contemporary of modern man? (1)
(Homo sapiens, Homo erectus, Homo habilis, Homo neanderthalensis)

2 Analyse the statements and choose the correct answer: (1)

Statement 1 : Night blindness and xerophthalmia are eye diseases caused by deficiency of Vitamin A.

Statement 2 : Prolonged deficiency of Vitamin A causes night blindness.

i) Statement 1 is true and 2 is false.

ii) Both statements are false.

iii) Statement 2 correctly explains statement 1.

iv) Both statements are true.

3. Match the different types of movements in **Column A** with the associated parts of the ear in **Column B** and choose the correct option: (1)

A Different types of movements	B Parts of the ear
P Linear movement of the head	1. Oval window
Q Rotational movement of the head	2. Hair cells of Organ of Corti
R Movement of endolymph in cochlea	3. Hair cells of vestibule
S Vibration of stapes	4. Round window
	5. Hair cells of semi circular canal

a) P-3, Q-2, R-5, S-4

b) P-3, Q-1, R-4, S-5

c) P-3, Q-5, R-2, S-1

d) P-2, Q-3, R-5, S-4

4. Choose the correct statements from the following: (1)

i. The **SRY** gene on the Y chromosome determines the development of the ovary in the embryo.

ii. Chromosomes that determine physical traits are autosomes.

iii. DNA strands wind around the histone octamer forms a structure called nucleosome

iv. In chromosome nucleosomes are connected by centromeres

a) i and ii true

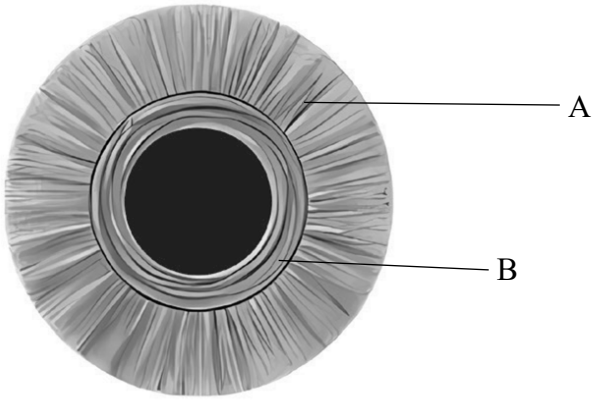
b) ii and iv true

c) ii and iii true

d) iii and iv true

Answer questions from 5 to 11 each carries 2 score **(7×2=14)**

5.A Observe the iluustration showing the muscles that control the size of the pupil and answer the questions.



i Identify the muscles A and B (1)

ii How does their action differ in bright and dim light . (1)

OR

5B. Analyse the statement and answer:

A variation in production of a neurotransmitter helps in distinguishing light and darkness.

i) Name the neurotransmitter. (1)

ii) How is darkness detected? (1)

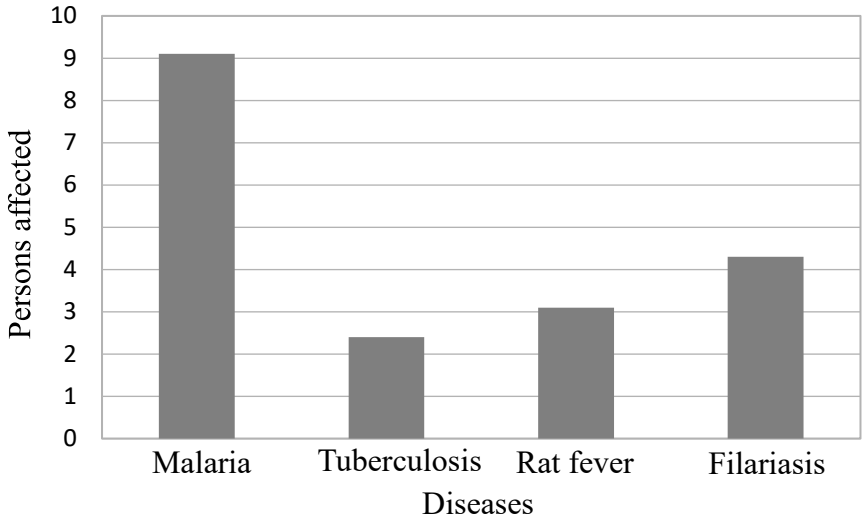
6. Read the news report and answer the questions

"DNA analysis of a fossil skull helped trace human ancestry."

i) Which technology made this possible? (1)

ii) What is the basis of this technology? (1)

7. Obsrve the graph and answer the questions



a) Which disease affected the largest number of people? (½)

b) Suggest precautions to prevent its spread. (1)

c) Identify the bacterial diseases from the graph. (½)

8. Some finch species found in the Galápagos Islands are listed in the box. (2)

Explain the formation of these species based on Darwin's theory of natural selection

insectivorous finches, cactus-eating finches, seed-eating finches.

9. Observe the figure and answer the questions (2)



If the RNA indicated becomes inactive, which stage of protein synthesis will be affected? How?

10. Analyse the table and answer the questions. (2)

Organism	**Difference in the amino acids of the beta chain as compared to humans**
Chimpanzee	0
Gorilla	1
Rat	31

a) Which branch of biology is this evidence given in the table related to? (½)

b) Explain the evolutionary relationship among these organisms based on the information? (1½)

11 A. A farmer's pepper plantation was visited by an agriculture officer and found that the plants are affected by quick wilt disease.

a) Identify the pathogen of this disease. (½)

b) What symptoms might have helped in identifying the disease? (1½)

OR

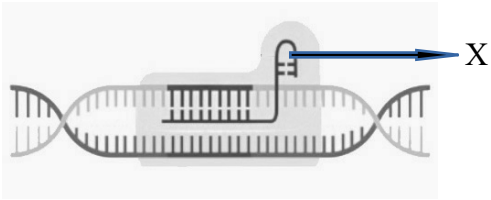
B. During a school blood donation camp, a student was given opportunity to match blood donors and recipients.

a) What factors would be considered for this? (1)

b) What happens if the donor's blood contains an antigen that is naturally absent in the recipient's blood? (1)

Answer questions from 12 to 17 each carries 3 scores **(6×3=18)**

12.

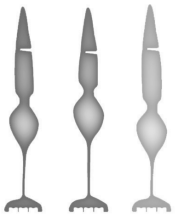


i) What does the illustration indicate? (1)

ii) Write the function of the part indicated as X (1)

iii) How does the process that occurs as a result of the action of the Cas9 enzyme bring about a change in the trait of an organism? (1)

13. Observe the illustration and answer the questions



i) Identify the type of cell shown. (1)

ii) What is the advantage of its sensitivity to different wavelengths? (2)

14 A. Read the statement and answer the questions

Currently, there is no medicine to completely cure this disease. However, through Anti Retro viral Therapy (ART), it is possible to control the multiplication of the virus and maintain the patient's immunity

i) Which disease is mentioned in the statement? How does it affect the body? (2)

ii) How can its transmission be prevented? (1)

OR

B. The box shows some proteins and enzymes involved in blood clotting: Analyse the information in the box and answer the questions.

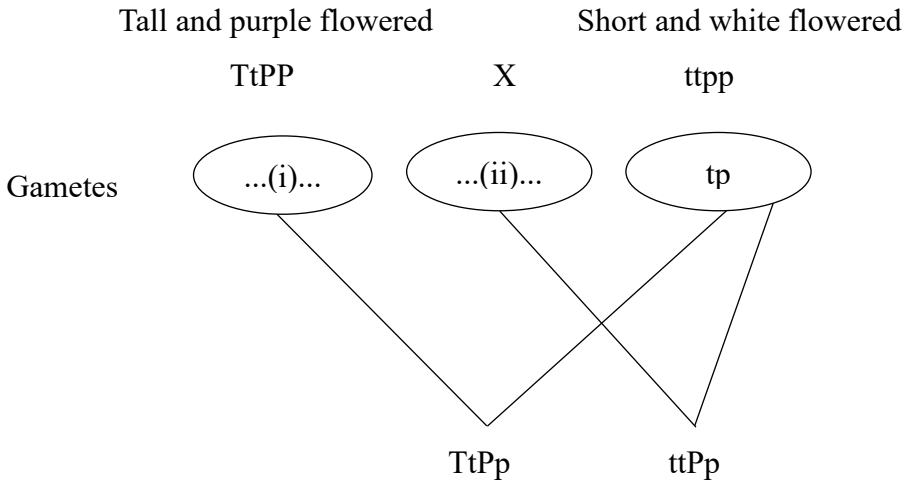
Prothrombin, Fibrinogen, Fibrin, Thromboplastin

a) Explain the process of blood clotting including the information in the sequential order. (2)

b) Beyond preventing blood loss, what is the importance of this process? (1)

15 A tall, purple-flowered pea plant (TtPP) was crossed with a short white-flowered plant (ttpp). Observe the illustration and answer the questions.

Parent Plants



i) Fill in the missing gametes ((i), (ii)) correctly (1)

ii) Did the offspring show any traits absent in parents? Explain it based on Mendel's hypothesis. (2)

16 A. Analyse the situations given below and answer the questions .

A – Calcium level in blood decreases.

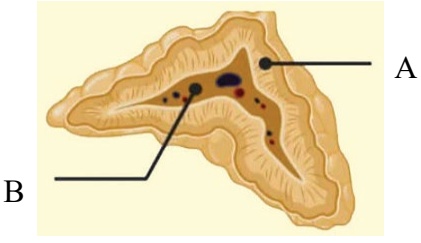
B – Calcium level in blood increases.

i) Which gland secretes the hormone required in situation B? (1)

ii) How does the hormone in situation A help maintain the normal level of calcium ? (2)

OR

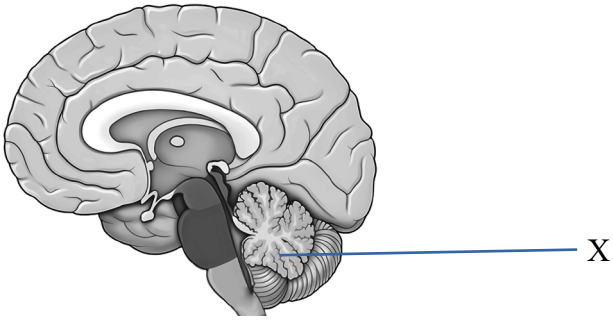
B. Observe the figure and answer the questions.



a) What are the hormones produced by A called? (1)

b) Mention any two hormones of this type and write two functions for each. (2)

17. Redraw the diagrams and anwer the questions according to the instructions.



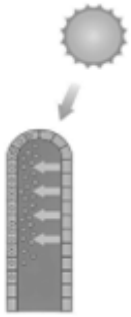
For redraw (1)

i) Label the part acting as an endocrine gland and state its function. (1)

ii) How does the part labelled X help to maintain body balance? (1)

Answer question 18. 4 Scores **(1×4=4)**

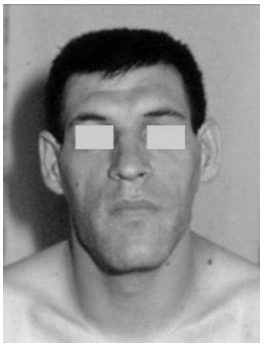
18 A. Analyse the diagram and answer the questions



i) The diagram shows movement of a hormone. Identify the hormone and the plant movement it causes. (1)

ii) How does this movement be possible? (3)

B. The figure shows a disorder related to the secretion of a tropic hormone.



a) Identify the hormone and name the hormone that controls its secretion. (1)

b) b) Mention disorders caused by its imbalance and their symptoms. (3)

Proforma for Scoring Key & Marking Scheme

SET C **BIOLOGY**

ANSWER KEY – ENGLISH MEDIUM

Item/ Qn. No.	Score	Answer/Value Points	Further Information
1	1	Homo neanderthalensis	
2	1	i. Statement 1 is true and 2 is false.	
3	1	P-3, Q-5, R-2, S-1	
4		c) ii, iii	
5A	1	A. Circular muscles contract B. Radial muscles contract	
	1	▪ In Intense light circular muscles contract. ▪ In dim light, Radial muscles muscles contract	
5B	1	i. Glutamate	
	1	▪ When glutamate is produced continuously, Onbipolar neurons become inactive and offbipolar neurons get activated. ▪ Signals indicating absence of light reach the brain	
6	1	DNA fingerprinting	
	1	ii. The sequence of nucleotides in each person vary. This uniqueness helps in identifying individuals.	
7		i. Malaria	
		ii. As the carriers of malarial parasites are mosquitoes, mosquito eradication	

		programmes and methods to control breeding of mosquitoes are to be practiced Tuberculosis, Rat fever	
8	1	Birds with beaks of suitable shape or size, depending on the availability of food resources in the environment, will survive..	
	1	They reproduce more effectively and create new generations. More variations accumulate over time leading to the creation of organisms that are unable to reproduce within the species. They evolve as new species.	
9	1	Translation	
	1	The inactive rRNA cannot add amino acids so protein synthesis stops, leading to metabolic disturbance	
10	1	Molecular biology	
	1	Humans are evolutionarily closest to chimpanzees. No difference is seen in the amino acid sequence of globulin in humans and chimpanzees. The next closest is gorilla (1 amino acid difference). The most distant relation is rat (31 amino acid differences).	
11A	1	Fungus	
	1	Wet marks on the leaves and stem. Leaves fall and the plant wilts quickly	
11B	1	Blood group and presence of Rh antigen is the primary factor in donor recipient matching	
	1	If the donor's blood contains an antigen that is naturally absent in the recipient's blood, the antibodies present in the recipient's	

		plasma may act against it, leading to agglutination of the blood	
12	1	CRISPR technology	
	2	The guide RNA precisely identifies the DNA segment that needs to be cut.	
	3	The sequence of nitrogen bases in DNA is edited so as to make changes in the characteristic of an organism. Through that, unwanted characteristics can be removed or new characteristics can be added.	
13	1	i. Cone cells	
	1	ii. There are three types of cone cells. S cone cells sensitivity at short wavelengths (blue light), M - cones at medium wavelengths (green light) and L - cones at longer wavelengths (red light). Shows higher sensitivity.	
	1	iii. Colour vision is made possible when the three types of cones get stimulated in varying proportions when exposed to coloured light depending upon the intensity and wavelength of light.	Expect open answers to this
14A	1	i. AIDS	
	1	ii. The virus attacks T lymphocytes. As a result, the body's ability to fight diseases is reduced	
	1	iii. - Avoid sexual contact with AIDS patients or take precautions - Avoid sharing of unsterilised needles and syringes, - Ensure proper testing while receiving blood and organs	

14B	**2**	i. When an injury occurs, the platelets and damaged tissues at the site produce an enzyme called thromboplastin. This converts a protein in the blood plasma called prothrombin into thrombin. Thrombin then converts an inactive protein called fibrinogen into fibrin. The fibrin forms a network of fibers that traps RBCs, forming a blood clot. This clot prevent further blood loss	
	1	ii. It blocks the entry of pathogens	
15		i. Tp,tP	
		ii. The trait dwarf purple (ttPp) is not a parental trait. When two or more different traits are combined, each trait is inherited independently to the next generation without mixing each other. (A pair of alleles in an organism does not influence the separation of another pair of alleles.)	
16.A	**1**	Thyroid gland.	
	1	▪ Parathormone ▪ Accelerates the reabsorption of calcium and phosphate to blood in kidneys. ▪ Stimulates the activity of mixing of calcium and phosphate from the bone into the blood. ▪ Accelerates the absorption of calcium from the intestine to blood	Any two points
16 B	**1**	i. Corticosteroids	
		Cortisol • Maintains glucose homeostasis • Suppresses the Immune responses	

		• Resists inflammation and allergy of the body	
	1	Aldosterone • Helps in maintaining equilibrium of minerals in the body • Regulates blood pressure Androgens • Controls sexual development and sexual behaviour	Any two hormones and it's any one function
17	**1**	For redrawing	
	2	i. Hypothalamus- Maintains homeostasis ii. Cerebellum - By coordinating muscular activities.	
18 A	**1**	i Auxin, phototropism	
	3	ii. Auxins have a tendency to stay away from light, When light falls on the stem from only one direction, auxin moves to the opposite side of the stem. There, auxin combines with hormone receptors in the cells to form a hormone receptor complex and as a result, changes occur in cellular activities. The cells on the shaded side elongate and the part where cell elongation occurs bends towards the opposite side.	
18B	**1**	i. Somatotropin Releasing hormone/Inhibitory hormone	
	3	Dwarfism - Stunted growth Gigantism - Excessive body growth Acromegaly - Excessive growth of body parts such as palms, foot and jaws	